



Vitamin B12 levels in plasma samples from the Parkinson's Progression Markers Initiative (PPMI)

PPMI Project ID: 243

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Summary

640 plasma samples from PPMI participants were available to determine the concentration of vitamin B12. Based on the smaller volumes of plasma samples available, we used a human vitamin B12 ELISA kit from Novus Biologicals (NBP2-60196) per manufacturer's instructions, as the detection method for the levels of vitamin B12.

Method

640 plasma samples from PPMI participants (with and without PD-pathogenic variants of *LRRK2* and *GBA*) were received from Denali Therapeutics (as remnants samples from MJFF Project # 180) on dry ice in a total of eight 384-well plates. Each plate was barcoded with a serial number and was referred to as a block. The volume for each sample in the plates was approximately 40µl.

For the present study, we only focused on the *LRRK2*+/*GBA*- and *LRRK2*-/*GBA*+ samples (n=465). The investigators running the assay were blinded to the genetic background of the plasma samples. A total of 15 assay runs were conducted to cover all the samples in 8 blocks. Before running each assay, the samples were thawed on wet ice. We chose 1:50 as the optimal dilution for these samples based on the results of several QC runs. 2.2 µl of the thawed samples were transferred from the respective wells of the 384-well plates into separate 0.5ml tubes and diluted in a 1:50 ratio in 107.8 µl of the sample dilution buffer provided in the kit. As per the kit manual, the required sample volume per well of the ELISA plate was 50µl, and since each sample was run in duplicates, we prepared a total of 110µl of each sample instead of 100µl to account for accidental pipetting losses.

A set of standards serially diluted (100 pg/ml to 0 pg/ml) in sample dilution buffer was also run in duplicates alongside the samples. We also ran pooled samples from healthy control and PD patients from each block in duplicates as internal controls for that particular block. Each 96-well ELISA plate could accommodate 8 standards, 2 pooled samples, and 38 plasma samples, all in duplicates.





Most blocks had more than 38 samples, so we divided the entire block into multiple ELISA runs, with each assay plate labeled as the block number followed by A, B, C, and so forth until all the samples in that block were covered. For example, we ran two ELISA plates for block 1 labeled 1A and 1B. Rather than splitting plates and exposing the remaining strips of wells of the 96-well ELISA plate precoated with the antibody against vitamin B12 to air for extended periods, some plates with samples from two different blocks were run with the respective pooled samples from both blocks.

The method followed to estimate the vitamin B12 levels through ELISA has been previously described by Li et al., 2019. Briefly, 50µl of 1X Biotin conjugate was added to each well of the plate containing 50µl of the standards and samples. The plates were sealed with an adhesive strip and incubated for 1h at 37°C, followed by 5 washes with the wash buffer provided with the kit. After completely removing all the wash buffer 100µl of 1X HRP-avidin was added to each well. The plate was sealed, incubated for 40 minutes at 37°C, and then washed five times. 90µl of TMB substrate was added to each well after the complete removal of the wash buffer, and the plate was incubated at 37°C for 20 minutes. Following this, 50ul of stop solution was added to each well and mixed with gentle tapping to allow color development. The plate was read at 450nm and 540nm in a plate reader within 5 minutes to record the absorbance of each well. Subtracting the absorbance obtained at 540nm from the absorbance at 450nm corrects for any optical imperfections in the plate. Therefore, all the calculations in this analysis were done on 540 nm wavelength corrected absorbance readings. A 4-parametric logistic (4-PL) standard curve was plotted for each ELISA run, which was used to calculate the concentration of vitamin B12 in each sample. The coefficient of variation (CV) was calculated for each sample pair, and the samples with CV > 0.20 were excluded from the final analysis. 44 samples (9.46%) were excluded due to a higher CV.

Each plate was batch-adjusted to its pooled samples to account for plate-to-plate variations. This considerably improved the overall CVs of the samples across the plates. For plate 8B, we used the smallest range of standards that fully straddled our sample values to obtain a better curve fit.

References

1. Li Z, Hardij J, Evers SS, Hutch CR, Choi SM, Shao Y, Learman BS, Lewis KT, Schill RL, Mori H, Bagchi DP, Romanelli SM, Kim KS, Bowers E, Griffin C, Seeley RJ, Singer K, Sandoval DA, Rosen CJ, MacDougald OA (2019). G-CSF partially mediates effects of sleeve gastrectomy on the bone marrow niche. *J Clin Invest.*, 129(6):2404-2416.

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